

An explicit modular approximation to $\zeta(7)$, with its exact asymptotics

River & Claude (Fable) — modular Apéry project, level 12, the same host as the $\zeta(5)$ page — 2 September 2026

The host is the one of the $\zeta(5)$ page: on $X_0(12)$ the Hauptmodul $h = \eta_1^3 \eta_4 \eta_6^2 / (\eta_2^2 \eta_3 \eta_{12}^3) = q^{-1} - 3 + 2q + \dots$ and the Fricke-invariant coordinate

$$x = \frac{h}{(h+3)(h+4)} = q - 4q^2 + 2q^3 + 12q^4 - 17q^5 - \dots, \quad x(i/\sqrt{12}) = x_+ = 7 - 4\sqrt{3},$$

with a fold at $\tau_* = i/\sqrt{12}$ and finite singularities only at $x = -1$ (the cusps $\frac{1}{2}, \frac{1}{6}$) and $x = 7 \mp 4\sqrt{3}$.

The source and its companions. The weight-8 Eisenstein combination $\Phi = \frac{1}{480}(E_8 - 572E_8(2\tau) + 11583E_8(3\tau) - 36608E_8(4\tau) + 46332E_8(6\tau) - 20736E_8(12\tau)) = q - 443q^2 + 13771q^3 - \dots$ has Dirichlet polynomial $P(s) = 1 - 572 \cdot 2^{-s} + 11583 \cdot 3^{-s} - 36608 \cdot 4^{-s} + 46332 \cdot 6^{-s} - 20736 \cdot 12^{-s}$ with $P(0) = P(2) = P(4) = P(6) = P(8) = 0$; since $L(\Phi, s) = P(s)\zeta(s)\zeta(s-7)$ and $\zeta(s-7)$ vanishes at $s = 3, 5$, every period except $\zeta(7)$ is annihilated, and $L(\Phi, 7) = P(7)\zeta(7)\zeta(0) = \frac{209}{1728}\zeta(7)$. Φ is Fricke-odd, so $\Phi(\tau_*) = 0$ and $U = \Phi/Dx$ is a holomorphic weight-6 form (Fricke-odd, $U(\tau_*) \neq 0$). Let $\Psi = \sum A_m m^{-7} q^m$ be the seventh Eichler integral ($\Phi = \sum A_m q^m$) and set

$$U' = (1+x) \frac{\Phi}{Dx} = \sum_{n \geq 0} c'_n x^n, \quad V' = U' \cdot \Psi = \sum_{n \geq 0} d'_n x^n$$

(the factor $1+x$ removes a simple pole of Φ/Dx at the cusp $x = -1$). Then $c'_n \in \mathbb{Z}$ and $D_n^7 d'_n \in \mathbb{Z}$ with $D_n = \text{lcm}(1, \dots, n)$, sharp:

$$c'_n : 1, -434, 8110, 68734, 581830, 5357278, 53776054, \dots \quad \frac{1728}{209} \frac{d'_n}{c'_n} : -\frac{864}{45353}, -\frac{1498041}{3389980}, \frac{1103599549}{1163597886}, \frac{497662108235}{504309763584}, \dots$$

Both sequences satisfy one explicit 19-term recurrence $\sum_{j=0}^{18} M_j(n) y_{n-j} = 0$ (resp. $= (-1)^n W(n)$, $\deg W = 6$, for d'_n) with $\deg M_j = 13$, $n^7 \mid M_0$, characteristic polynomial $(\lambda+1)^6(\lambda^2 - 14\lambda + 1)^6$; c'_n alone satisfies a 13-term recurrence with coefficients of degree 49 and characteristic polynomial $(\lambda+1)^6(\lambda^2 - 14\lambda + 1)^3$, the fewest terms possible. The coefficients run to hundreds of digits (the order-7 minimal operator carries an apparent-singularity divisor of degree 42) and are kept in `lattice/zeta7_showcase/`.

Theorem (limit, rate, and all three constants).

$$\frac{1728}{209} \frac{d'_n}{c'_n} \longrightarrow \zeta(7), \quad \frac{d'_n}{c'_n} - \frac{209}{1728} \zeta(7) \sim \frac{14161}{720 K'_7} (-1)^{n+1} (7 - 4\sqrt{3})^n n^{1/2} (\log n)^6, \quad c'_n \sim K'_7 (7 + 4\sqrt{3})^n n^{-3/2},$$

$$K'_7 = (8 - 4\sqrt{3}) \frac{3^{27/4} (739 - 356\sqrt{3})}{2^{77/6}} \cdot \frac{\Gamma(1/3)^{12}}{\pi^{19/2}} = 77.239145578434321892713944136629364 \dots$$

Why. Fricke invariance makes $V' - \frac{209}{1728} \zeta(7) U'$ regular at the fold, so the linear form is driven by the cusp $x = -1$, where it behaves as $\alpha' \log^7(1+x)$. The new cusp-constant theorem gives α' as a product of two constant terms at that cusp (width $w = 3$): $\alpha' = \varphi_0 a_0 w^7 / 7!$ with $a_0(\Phi) = -\frac{119}{81}$ and $\varphi_0(U') = -\frac{119}{27}$, i.e.

$$\alpha' = \frac{119^2}{7!} = \frac{14161}{5040}, \quad d'_n - \frac{209}{1728} \zeta(7) c'_n \sim \frac{14161}{720} (-1)^{n+1} \frac{(\log n)^6}{n}$$

(the same theorem returns the $\frac{13}{480}$ of the $\zeta(5)$ page; verified here by direct evaluation near the cusp to 8 digits). The fold constant comes from $K_7'^2 = x_+(1+x_+)^2 (DU)(\tau_*)^2 / (2\pi(-D^2x)(\tau_*))$ with the differentiated Fricke law $DU(\tau_*) = \frac{6\sqrt{12}}{4\pi} U(\tau_*)$: $U(\tau_*)$ is a genuine weight-6 CM value, made exact by the identity $v^3 U / (Dv)^3 = \mathcal{N}(v) / ((1+2v)^4 (1+3v)^2 (1+4v)^2 (1+6v)^4)$, $v = 1/h$, $\mathcal{N}$ of degree 12, giving $U(\tau_*) / Dv(\tau_*)^3 = 17123724 - 9885726\sqrt{3}$; the transcendental part is the same CM period as for $\zeta(5)$, $(Dv(\tau_*) / \Omega^2)^3 = 3(45 + 26\sqrt{3}) / 2^{17}$, $\Omega^2 = \Gamma(1/3)^6 / \pi^4$. The number $K'_7 \pi^{19/2} / \Gamma(1/3)^{12}$ has degree 12 over $\mathbb{Q}$; the closed form agrees with the fold formula to 700 digits and with Richardson extrapolation of the exact c'_n ($n \leq 3000$) to 91 digits.

Numbers. Exact c'_n, d'_n to $n = 3000$ from the recurrence, 3600-digit arithmetic:

n	digits of $\zeta(7)$ in $\frac{1728}{209} \frac{d'_n}{c'_n}$	$ d'_n/c'_n - \frac{209}{1728} \zeta(7) ^{1/n}$	error / leading asymptotic
5	2.25	0.2322	35.8
20	17.9	0.1149	14.8
100	108.2	0.0811	8.31
400	450.5	0.0744	5.95
900	1022.0	0.0730	5.12
3000	3423.6	0.0722	4.26

The n -th root of the error tends to $7 - 4\sqrt{3} = 0.07180$; the ratio to the leading term tends to 1 only logarithmically.

What it does and does not do. The rate $\log \frac{1}{7-4\sqrt{3}} = 2.634$ against $\log D_n^7 \sim 7n$ gives no irrationality proof ($2.63 < 7$; Apéry's $\zeta(3)$ form has $3.53 > 3$). What is new is a $\zeta(7)$ system as explicit as the $\zeta(5)$ one, on the same host with the same CM period, differing only in the weight of the source; the project's earlier level-12 construction, on the coordinate $t = x' / (16x'^2 + 2x' + 1)$ with $x' = \eta_4^2 \eta_{12}^2 / (\eta_1^2 \eta_3^2)$, has a non-modular branch point at $t = -\frac{1}{6}$ and converges only at rate 0.6 per term. Files: `consolidation/ASYMPTOTIC.CONSTANTS.md` (§5, the cusp-constant and Casoratian theorems), `lattice/zeta7_showcase/`.